

Sizhu Lu

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Homepage: <https://sizhu-lu.github.io>

Research interests Causal inference, Semiparametric theory, Clinical trials, AI in causal inference

Academic appointments **Stanford University** Palo Alto, CA
Postdoctoral Fellow, Stanford Data Science Institution 06/2026 – 06/2027

Carnegie Mellon University Pittsburgh, PA
Assistant Professor of Statistics & Data Science 08/2027 –

Education **University of California, Berkeley** Berkeley, CA
Ph.D. candidate in Statistics 05/2026
Advisor: Professor Peng Ding

University of California, Berkeley Berkeley, CA
Ph.D. in Business Administration, Haas School of Business 08/2022
Advisor: Professor Amir Kermani
Major: Finance and Real Estate

Peking University Beijing, China
B.S. in Mathematics and Applied Mathematics & B.A. in Finance 07/2017

Harvard University Cambridge, MA
Visiting undergraduate student Fall 2016

Publications and manuscripts **Principal stratification with continuous post-treatment variables: non-parametric identification and semiparametric estimation**
Sizhu Lu, Zhichao Jiang, and Peng Ding, 2026, *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 88, 239–260

The roles of estimands and assumptions in causal inference: Comment on “Chasing shadows: how implausible assumptions skew our understanding of causal estimands”
Peng Ding and **Sizhu Lu**, 2025, *Statistics in Biopharmaceutical Research*, 17, 523–529

Estimating within-cluster and between-cluster spillover effects in randomized saturation designs
Sizhu Lu^{*}, Lei Shi^{*}, and Peng Ding, 2026, *Social Networks*, 87, 24–34

Efficient inference for noisy LLM-as-a-judge evaluation

Yiqun T Chen, **Sizhu Lu**, Sijia Li, Moran Guo, and Shengyi Li, 2026, [International Conference on Machine Learning \(ICML\)](#)

Estimating treatment effects with competing intercurrent events in randomized controlled trials

Sizhu Lu, Yanyao Yi, Yongming Qu, Karen Liu, Ting Ye, and Peng Ding, 2025+, *Major revision at Journal of the American Statistical Association*, [arXiv preprint arXiv:2503.03049](#)

Design-based causal inference in bipartite experiments

Sizhu Lu^{*}, Lei Shi^{*}, Yue Fang, Wenxin Zhang, and Peng Ding, 2025+, *under review*, [arXiv preprint arXiv:2501.09844](#)

Flexible sensitivity analysis for causal inference in observational studies subject to unmeasured confounding

Sizhu Lu and Peng Ding, 2023+, *under review*, [arXiv preprint arXiv:2305.17643](#)

TERRA: A transformer-enabled recursive R-learner for longitudinal heterogeneous treatment effect estimation

Lei Shi, **Sizhu Lu**, Rita Qiuran Lyu, Peng Ding, and Nikos Vlassis, 2025+, *under review*, [arXiv preprint arXiv:2510.22407](#)

GAUGER: Generalized regression adjustment via graph-weighted exposure-level residualization

Lei Shi, Rita Qiuran Lyu, and **Sizhu Lu**, 2026+, *under review*

Selected research in progress

Combining treatment policy and hypothetical strategies for semi-competing intercurrent events

joint work with Yanyao Yi, Yongming Qu, Ting Ye, and Peng Ding

Simple yet efficient weighting estimation for the marginal quantile treatment effect in randomized experiments

joint work with Peng Ding

Two-phase sampling for the local average treatment effect: efficient estimation and optimal design

joint work with Peng Ding

Instrumental variable methods are widely used for causal inference with unmeasured confounding, yet practical applications often involve partial data collection due to cost or design constraints. Motivated by modern large-scale randomized experiments with noncompliance and outcomes that are costly to measure, we study the identification and estimation of the local average treatment effect under two-phase sampling. We first show that the canonical two-stage least squares estimator must be weighted properly, and that the corresponding standard error must be modified according to the two-phase sampling design. We then propose a semiparametrically efficient and multiply robust estimator based on the efficient influence function. More importantly, we argue that two-phase sampling can utilize post-treatment variables that are predictive of the outcome, and discuss the corresponding estimation and inference problems. Our theory not only provides the basis for the optimal two-phase sampling but also covers the setting of missing outcomes under the local average treatment effect framework.

Honors and scholarships	ASA Student Paper Award in the Biopharmaceutical Section	2026
	NSF Travel Award for the ICSA 2026 Applied Statistics Symposium	2026
	KAG Travel Grant, UC Berkeley	2026
	Graduate Division Conference Travel Grant, UC Berkeley	2024, 2026
	San Francisco ASA JSM Travel Award	2024
	Society for Political Methodology Polmeth Meeting Travel Award	2024
	Outstanding Graduate Student Instructor	2023
	Outstanding Graduate of Beijing (Top 1%)	2017
	Merit Student of Peking University (Top 2%)	2014 – 2016
Teaching experience	Graduate Student Instructor, Department of Statistics, UC Berkeley	
	STAT 135: Concepts of Statistics	Spring 2025
	STAT 215B: Statistical Models: Theory and Application	Spring 2024
	STAT 156/256: Causal Inference	Fall 2022, Fall 2023
	STAT 230A: Linear Models	Spring 2021, Spring 2023
	Guest Lecturer, Department of Statistics, UC Berkeley	
	STAT 156/256: Causal Inference	10/2025, 10/2022
	STAT 230A: Linear Models	04/2023
	Graduate Student Instructor, Haas School of Business, UC Berkeley	
	MFE 230M: Asset-Backed Security Markets	Fall 2020 – 2022
	MFE 230I: Fixed Income Markets	Summer 2020
	MBA 283: Real Estate Finance and Securitization	Fall 2019
	UGBA 180: Introduction to Real Estate & Urban Land Economics	Spring 2019
	Instructor, Haas School of Business, UC Berkeley	
	Finance Net Present Value, Business Academy for Youth	Summer 2019

Talks and posters	Talk at 2026 INFORMS, Invited Paper Session	11/2026
	Talk at 2026 JSM Contributed Paper Session	08/2026
	Talk at 2026 Applied Statistics Symposium, ICSA	06/2026
	Talk at the SILVER lab, Department of Epidemiology and Public Health, University of Maryland School of Medicine	06/2026
	Poster at 2026 American Causal Inference Conference	05/2026
	Talk at the IROM Department, UT Austin McCombs	02/2026
	Talk at the Department of Statistics and Data Science, UPenn Wharton	02/2026
	Talk at the Department of Statistics and Data Science, CMU	02/2026
	Talk at the Econometrics and Statistics Colloquium, Chicago Booth	01/2026
	Talk at the ASA Biopharmaceutical Webinar	11/2025
	Poster at 2025 CLIMB Retreat, UC Berkeley	11/2025
	Talk at Online Causal Inference Seminar	09/2025
	Poster at 2025 Berkeley Statistics Annual Research Symposium	09/2025
	Talk at the Emory Causal Network Analysis Workshop	08/2025
	Poster at Berkeley Experimental Designs in the Era of Artificial Intelligence Workshop	03/2025
	Talk at 2024 INFORMS, Invited Paper Session	10/2024
	Poster at 2024 Stanford Berkeley Joint Colloquium	10/2024
	Talk at 2024 JSM Topic-Contributed Paper Session	08/2024
	Poster at 2024 PolMeth	07/2024
	Talk at School of Management and Economics, CUHK	05/2024
	Poster at 2024 American Causal Inference Conference	05/2024
	Talk at Eli Lilly and Company (G4S Seminar)	05/2024
	Talk at 2024 Berkeley Statistics Annual Research Symposium	04/2024
	Poster at Winter Workshop: Causal Inference and Its Applications, University of Florida	01/2024
	Poster at 2023 CLIMB Retreat, UC Berkeley	11/2023
	Talk at Center for Statistical Science, Tsinghua University	06/2023
Reviewing service	Journal of the American Statistical Association, Annals of Statistics, Biometrika, Journal of the Royal Statistical Society, Series B, Journal of Causal Inference, Statistica Sinica, Journal of Educational and Behavioral Statistics, Biometrical Journal, Biostatistics, Journal of Computational and Graphical Statistics, Sociological Methods and Research, Transactions on Machine Learning Research, Social Networks.	
Statistical consulting	National Security Agency	
	<i>Science Advisory Group</i>	2024, 2025
	• Selected as part of a Berkeley-led advisory group that provides consulting on statistical problems posed by the National Security Agency.	

- Analyzed problem statements derived from real-world intelligence and security contexts (with sensitive details removed), discussed solutions through a series of technical meetings, and prepared written reports applying statistical theory.

Eli Lilly and Company

Academic Contractor: Consulting Statistician

09/2024 – 03/2026

Statistician Co-op Intern

01/2024 – 05/2024

- Collaborated with Eli Lilly statisticians to address methodological challenges in the analysis of randomized controlled trials involving intercurrent events.
- Developed causal estimands and estimators that bridge regulatory estimand strategies with practical trial analysis.
- This collaboration directly led to my papers “*Estimating treatment effects with competing intercurrent events in randomized controlled trials*” and “*Combining treatment policy and hypothetical strategies for competing intercurrent events*”.

Industry experience

Netflix

Los Gatos, CA

Experimentation and Causal Inference Intern

05/2023 – 08/2023

- Conducted causal analyses of business decisions using longitudinal observational data.
- Applied structural nested mean models under parallel-trends assumptions to estimate dynamic treatment effects.
- This work deepened my understanding of causal inference in longitudinal settings and directly informed the later conference paper “*Transformer-Enabled Recursive R-Learner (TERRA)*”, integrating causal structure with modern machine learning architectures.

Programming skills

Proficient in: R, Python, MATLAB, and Stata.